



## **The MFU AP-MAP analysis system**

**Watcharaphon Somsuk**  
**Suwatchaphong Ketsora**  
**Patcharaporn Chamniwikaiwet**

**BACHELOR OF ENGINEERING  
IN COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY**

**2023**

**© COPYRIGHT BY MAE FAH LUANG UNIVERSITY**

# **The MFU AP-MAP analysis system**

**Watcharaphon Somsuk**

**Suwatchaphong Ketsora**

**Patcharaporn Chamniwikaiwet**

A COMPUTER ENGINEERING PROJECT SUBMITTED TO  
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF  
THE REQUIREMENTS FOR THE DEGREE OF  
BACHELOR OF ENGINEERING  
IN COMPUTER ENGINEERING

**MAE FAH LUANG UNIVERSITY**

**2023**

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

## **The MFU AP-MAP analysis system**

|                      |                       |
|----------------------|-----------------------|
| <b>Watcharaphon</b>  | <b>Somsuk</b>         |
| <b>Suwatchaphong</b> | <b>Ketsora</b>        |
| <b>Pacharaporn</b>   | <b>Chamniwikaiwet</b> |

THIS COMPUTER ENGINEERING PROJECT HAS BEEN  
APPROVED TO BE A PARTIAL FULFILLMENT OF THE  
REQUIREMENTS FOR THE DEGREE OF BACHELOR OF  
ENGINEERING IN COMPUTER ENGINEERING 2022

### EXAMINING COMMITTEE

|                                                                                      |                                                                                      |                |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------|
|   |  | .....ADVISOR   |
| (Dr.Mahamah Sebakor)                                                                 |                                                                                      |                |
|  |                                                                                      | .....COMMITTEE |
| (Asst.Prof.Suppakarn Chansareewittaya)                                               |                                                                                      |                |
|   |                                                                                      | .....COMMITTEE |
| (Asst.Prof.Gp.Capt.Dr.Thongchai Yooyativong)                                         |                                                                                      |                |

## **ACKNOWLEDGEMENT**

Foremost, we would like to express our sincere gratitude to our precious advisor Asst. Aj. Dr.Mahamah Sebakor for supporting our project and giving the opportunity for us to do our research and developing this project. Without valuable guidance, this project might not succeed.

With all support from our parents with encouragement and fortune propelling us to our success we were very grateful and we felt very appreciative.

Lastly, we would like to thank any committee and any suggestion from everyone for helping us improve the quality of our work and making this project in the best state possible.

|               |                |
|---------------|----------------|
| Watcharaphon  | Somsuk         |
| Suwatchaphong | Ketsora        |
| Patcharaphorn | Chamniwikaiwet |

|                                  |                                                   |           |
|----------------------------------|---------------------------------------------------|-----------|
| <b>Title</b>                     | The MFU AP-MAP analysis system                    |           |
| <b>Author</b>                    | Miss Watcharaphon Somsuk                          |           |
|                                  | Mr. Suwatchaphong Ketsora                         |           |
|                                  | Miss Patcharaporn Chamniwikaiwet                  |           |
| <b>Degree</b>                    | Bachelor of Engineering<br>(Computer Engineering) |           |
| <b>Supervisory<br/>Committee</b> | Dr.Mahamah Sebakor                                | Advisor   |
|                                  | Asst.Prof.Suppakarn Chansareewittaya              | Committee |
|                                  | Asst.Prof.Gp.Capt.Dr.Thongchai Yooyativong        | Committee |

## Abstract

This computer engineering project focuses on analyzing Mae Fah Luang University's wireless connectivity system, with a primary goal of identifying blind spots crucial for network administrators. Ensuring complete coverage and user satisfaction within the wireless network is imperative. The system utilizes specialized software to collect and analyze data, effectively pinpointing areas with inadequate or absent signals recognized as blind spots.

Developed by the School of Information Technology, this system offers a valuable tool empowering network administrators to optimize wireless coverage and enhance user experiences significantly.

# CONTENTS

|                                           |           |
|-------------------------------------------|-----------|
| <b>ACKNOWLEDGEMENT.....</b>               | <b>2</b>  |
| <b>Abstract.....</b>                      | <b>3</b>  |
| <b>CONTENTS.....</b>                      | <b>4</b>  |
| <b>LIST OF TABLES.....</b>                | <b>5</b>  |
| <b>LIST OF FIGURES.....</b>               | <b>6</b>  |
| <b>CHAPTER 1</b>                          |           |
| <b>INTRODUCTION.....</b>                  | <b>7</b>  |
| 1.1 Background and Rationale.....         | 7         |
| 1.2 Objective.....                        | 7         |
| 1.3 Scope.....                            | 7         |
| 1.4 Methodology.....                      | 8         |
| <b>1.5 Plan.....</b>                      | <b>9</b>  |
| 1.6 Expected Result.....                  | 10        |
| 1.7 Place and Equipment.....              | 10        |
| <b>CHAPTER 2 LITERATURE REVIEW.....</b>   | <b>11</b> |
| 2.1 Related Theory and Research.....      | 11        |
| 2.2 Related Work.....                     | 12        |
| 2.3 Related Technology.....               | 13        |
| <b>CHAPTER 3 ANALYSIS AND DESIGN.....</b> | <b>18</b> |
| 3.1 Existing System.....                  | 18        |
| 3.2 New System.....                       | 19        |
| 3.3 Use case diagram.....                 | 20        |
| 3.4 Activity diagram.....                 | 21        |
| <b>CHAPTER 4 CONCLUSION.....</b>          | <b>22</b> |
| 4.1 Conclusion.....                       | 22        |
| 4.2 Discussion.....                       | 22        |
| <b>REFERENCES.....</b>                    | <b>23</b> |

## **LIST OF TABLES**

|                             |   |
|-----------------------------|---|
| Table 1.1 Working plan..... | 9 |
|-----------------------------|---|

## LIST OF FIGURES

|                                               |    |
|-----------------------------------------------|----|
| Figure 1 Access Point.....                    | 13 |
| Figure 2 Router.....                          | 13 |
| Figure 3 Laptops, Computers, and Tablets..... | 14 |
| Figure 4 Visual Studio Code.....              | 14 |
| Figure 5 Python.....                          | 15 |
| Figure 6 Scikit-Learn.....                    | 15 |
| Figure 7 Google Map API.....                  | 16 |
| Figure 8 PuTTY.....                           | 16 |
| Figure 9 VMware Workstation.....              | 17 |
| Figure 10 Existing System.....                | 18 |
| Figure 11 New System.....                     | 19 |
| Figure 12 Use case diagram.....               | 20 |



# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 Background and Rationale**

Mae Fah Luang University (MFU) is a large campus with many buildings, and wireless connectivity is essential for students, faculty, and staff to access online resources and communicate with each other. However, due to the complex nature of wireless signals, there may be areas on the campus where wireless signals are weak or non-existent, also known as dead spots.

Analyzing wireless signals to identify dead spots is an important task for network administrators to ensure that wireless coverage is adequate and meets users' needs. Therefore, MFU's School of Information Technology has developed a system to analyze wireless signals and identify dead spots on the campus. The system works by using software to collect data and analyze it to identify areas with weak or non-existent signals, which will be marked as a dead point.

In summary, analyzing wireless signals to identify dead spots is an important task in ensuring a reliable wireless network on the MFU campus. The system developed by the School of Information Technology provides a valuable tool for network administrators to optimize wireless coverage and improve the user experience.

### **1.2 Objective**

1.2.1 The access point analysis that should be added analyzes the feature plan for the WLC network at Mae Fah Luang University.

### **1.3 Scope**

1.3.1 Analyzing access points using state in Mae Fah Luang University.

## **1.4 Methodology**

- 1.4.1 Proposal preparation and defense
- 1.4.2 Literature review and requirement gathering
- 1.4.3 System analysis
- 1.4.4 System design
- 1.4.5 Senior project 1 document preparation
- 1.4.6 Senior Project 1 exam
- 1.4.7 System development
- 1.4.8 System testing
- 1.4.9 Senior Project 2 document preparation exam
- 1.4.10 Senior Project 2 exam

## 1.5 Plan

### Table 1.1 Working plan

|                                             | Senior project 1 |     |     |     |     | Senior project 2 |     |     |     |     |
|---------------------------------------------|------------------|-----|-----|-----|-----|------------------|-----|-----|-----|-----|
| <b>Plan/Month</b>                           | Jan              | Feb | Mar | Apr | May | Aug              | Sep | Oct | Nov | Dec |
| Proposal preparation and defense            |                  |     |     |     |     |                  |     |     |     |     |
| Literature review and requirement gathering |                  |     |     |     |     |                  |     |     |     |     |
| System analysis                             |                  |     |     |     |     |                  |     |     |     |     |
| System design                               |                  |     |     |     |     |                  |     |     |     |     |
| Senior project 1 document preparation       |                  |     |     |     |     |                  |     |     |     |     |
| Senior Project 1 exam                       |                  |     |     |     |     |                  |     |     |     |     |
| System development                          |                  |     |     |     |     |                  |     |     |     |     |
| System testing                              |                  |     |     |     |     |                  |     |     |     |     |
| Senior project 2 document preparation       |                  |     |     |     |     |                  |     |     |     |     |
| Senior Project 2 exam                       |                  |     |     |     |     |                  |     |     |     |     |

## **1.6 Expected Result**

1.6.1 Analyzing user data for each wireless access point entails the systematic collection

1.6.2 Examination of usage patterns and behaviors specific to individual access points within a network.

## **1.7 Place and Equipment**

1.7.1 Place

(1) Access point in Mae Fah Luang University

1.7.2 Hardware

(1) Access point

(2) Router

(3) Laptop, Computer and Tablets

1.7.3 Software

(1) Visual Studio Code

(2) Python

(3) Scikit-Learn

(4) Google Map APIs

(5) PUTTY

(6) VMware Workstation

## **CHAPTER 2**

### **LITERATURE REVIEW**

#### **2.1 Related Theory and Research**

##### **2.1.1 Coverage and Capacity Planning Theory**

Coverage and Capacity Planning Theory serves as a foundational framework for network planners and administrators to strategically deploy access points, ensuring comprehensive coverage, adequate capacity, and optimal network performance based on thorough analysis and estimations of coverage area, user density, and traffic demands.

##### **2.1.2 Signal Propagation Models**

Signal Propagation Models are instrumental in predicting signal behavior, allowing network planners to anticipate areas with potential signal weaknesses. These models help determine optimal access point placement and configurations, ensuring adequate signal coverage and strength for reliable wireless communication within a network.

##### **2.1.3 Spatial Analysis And GIS**

Spatial Analysis and GIS Techniques offer valuable insights into the geographical aspects of a network. By visualizing and analyzing spatial data, network planners gain a better understanding of user distributions and coverage patterns, aiding in strategic decisions regarding access point deployment, optimizing network coverage, and ensuring efficient wireless communication across different geographic areas within the network.

##### **2.1.4 Traffic Engineering and Network Optimization**

Traffic Engineering and Network Optimization theories help in understanding and managing traffic demands within a network infrastructure. By estimating traffic patterns, optimizing resource allocation, and ensuring efficient traffic flow, these theories contribute to effective network planning, especially in determining the number and placement of access points to handle expected traffic demands effectively.

##### **2.1.5 Machine Learning and Predictive Analytics**

Machine Learning and Predictive Analytics provide valuable insights into future network demands and user behavior. By leveraging historical data and advanced algorithms, these techniques assist in making informed decisions regarding access point deployments, optimizing network resources, and ensuring the network's ability to meet evolving user needs effectively.

These references and theories encompass various aspects of network planning, signal analysis, user behavior, and predictive modeling, providing insights into determining the number of access points needed and identifying areas where additional access points should be deployed. Consulting these resources can offer valuable insights into theoretical frameworks supporting data analysis for access point usage and deployment in networks.

## **2.2 Related Work**

### **2.2.1 System for User Signal Distribution Analysis**

Research Outcome: Development of a system or methodology using data analytics, machine learning, or spatial analysis to analyze user signal distribution across an area.

References:

Specific research articles or papers might include works discussing methodologies or systems designed for user signal analysis in wireless networks. Consider looking for papers in journals such as IEEE Transactions on Wireless Communications or conferences like ACM MobiCom.

### **2.2.2 Access Point Deployment Optimization for Coverage Improvement**

Research Outcome: Proposing methodologies or algorithms to optimize access point deployment in areas with poor coverage based on user signal distribution analysis.

References:

Research papers might exist in journals like IEEE Transactions on Mobile Computing or conferences such as IEEE INFOCOM, discussing approaches to optimize access point placement for coverage improvement.

### **2.2.3 Spatial Analysis and GIS-Based Solutions**

Research Outcome: Using GIS or spatial analysis techniques to visualize user signal distribution and identify regions with inadequate coverage.

References:

Explore articles in journals like Computers, Environment and Urban Systems or International Journal of Geographical Information Science that might present GIS-based solutions for access point placement in areas with poor coverage.

### **2.2.4 Machine Learning for Access Point Optimization**

Research Outcome: Utilizing machine learning models to predict signal coverage patterns and suggest optimal access point locations.

References:

Look for articles in journals such as IEEE Wireless Communications and Networking Conference (WCNC) proceedings or IEEE Transactions on Vehicular Technology that discuss machine learning-based approaches for access point optimization.

To find specific research studies or systems focusing on user signal distribution analysis and access point deployment in areas with poor coverage, it's recommended to search academic databases like IEEE Xplore, ACM Digital Library, Google Scholar, or ScienceDirect using relevant keywords such as "user signal distribution analysis," "access point deployment optimization," "wireless network coverage improvement," or "GIS-based access point planning." These databases often contain scholarly articles, papers, and conference proceedings relevant to these research areas.

## 2.3 Related Technology

### 2.3.1 Hardware

#### (1) Access Point

Access Point (AP) is a device that enables wireless devices to connect to a wired network using Wi-Fi. It acts as a central hub for wireless communication, allowing devices such as laptops, smartphones, tablets, and other Wi-Fi-enabled devices to connect to a local area network (LAN) without requiring physical wired connections.

The primary function of an access point is to create a wireless local area network by broadcasting a Wi-Fi signal, allowing devices within its range to establish a connection and access resources on the network or the internet. Access points are commonly used in homes, businesses, public spaces, and various other environments to provide wireless connectivity.



Figure 1 Access Point

#### (2) Router

A router is a networking device that forwards data packets between computer networks. It serves as a central point that enables communication between devices within a local network (LAN) or between multiple networks, such as connecting a home network to the internet.



Figure 2 Router

### (3) Laptops, Computers, and Tablets

Laptops, computers, and tablets are all commonly used devices to access information or services via the internet, utilizing either a local area network (LAN) connection or a wireless connection.

Laptops and Computers are both powerful computing devices used to access the internet via LAN or wireless connections.

Tablets are portable devices equipped with touchscreens, often used for accessing the internet and online services. Similar to laptops, tablets can connect to the internet using Wi-Fi connectivity, enabling wireless access to the internet via Wi-Fi routers or access points.



Figure 3 Laptops, Computers, and Tablets

## 2.3.2 Softwares

### (1) Visual Studio Code

Visual Studio Code is a streamlined code editor with support for development operations like debugging, task running, and version control. It aims to provide just the tools a developer needs for a quick code-build-debug cycle and leaves more complex workflows to fuller featured IDEs, such as Visual Studio IDE.



Figure 4 Visual Studio Code



## (2) Python

Python is a computer programming language often used to build websites and software, automate tasks, and conduct data analysis. Python is a general-purpose language, meaning it can be used to create a variety of different programs and isn't specialized for any specific problems. In Python, object-oriented Programming (OOPs) is a programming paradigm that uses objects and classes in programming. It aims to implement real-world entities like inheritance, polymorphisms, encapsulation, etc. in the programming.



Figure 5 Python

## (3) Scikit-Learn

Scikit-learn is a versatile and widely used open-source machine-learning library for Python. It provides a comprehensive suite of tools and algorithms that facilitate various machine-learning tasks, including classification, regression, clustering, dimensionality reduction, and more.



Figure 6 Scikit-Learn

#### (4) Google Map API

The Google Maps Api is one of those clever bits of Google technology that helps you take the power of Google Maps and put it directly on your own site. It lets you add relevant content that is useful to your visitors and customize the look and feel of the map to fit with the style of your site.



Figure 7 Google Map API

#### (5) PuTTY

PuTTY is a free, open-source, and highly popular terminal emulator program that supports various network protocols, including SSH (Secure Shell), Telnet, login, and serial connections. It is primarily used on Windows platforms to establish secure connections to remote machines or network devices.

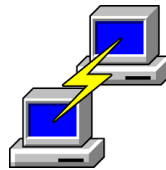


Figure 8 PuTTY

#### (6) VMware Workstation

VMware caters to various needs, including server virtualization, desktop virtualization, network virtualization, storage solutions, and cloud management. They are widely used by businesses, enterprises, IT professionals, developers, and individuals to create, manage, and optimize virtualized environments and cloud infrastructure.



Figure 9 VMware Workstation

## CHAPTER 3

### ANALYSIS AND DESIGN

#### 3.1 Existing System

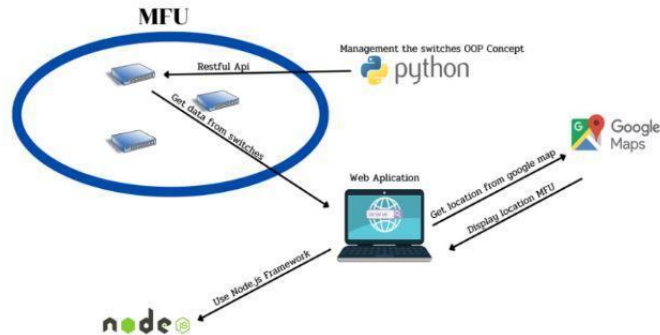


Figure 10 Existing System

With the current system, we have data available to display the status and user count of access points. However, we are unable to specify which specific access points need to be installed or deactivated at any given time to better accommodate users. This situation necessitates us to make guesses, which may lead to challenges in asset allocation and potential issues, such as slow internet speed, if there is an excessive number of users. The lack of precise information on when and where additional access points are needed or when deactivation is required poses a challenge in optimizing the system to better meet user needs. This uncertainty may result in difficulties in asset allocation and potential problems, such as slow internet speeds, when the number of users is high.

## 3.2 New System

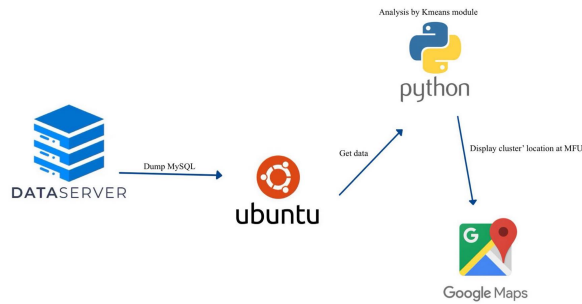


Figure 11 New System

After extracting data from the wifimap2000 network clone for storage in Ubuntu, we proceeded to conduct analysis using Python and the scikit-learn library, leveraging the KMeans module. The primary objective of this analysis is to identify locations suitable for the installation of additional access points. The results are visually presented through graphs, providing in-depth information about the latitude and longitude coordinates conducive to strategically placing new access points. Additionally, the outcomes are showcased on Google Maps, enhancing spatial context and user-friendly presentation.

### 3.3 Use case diagram

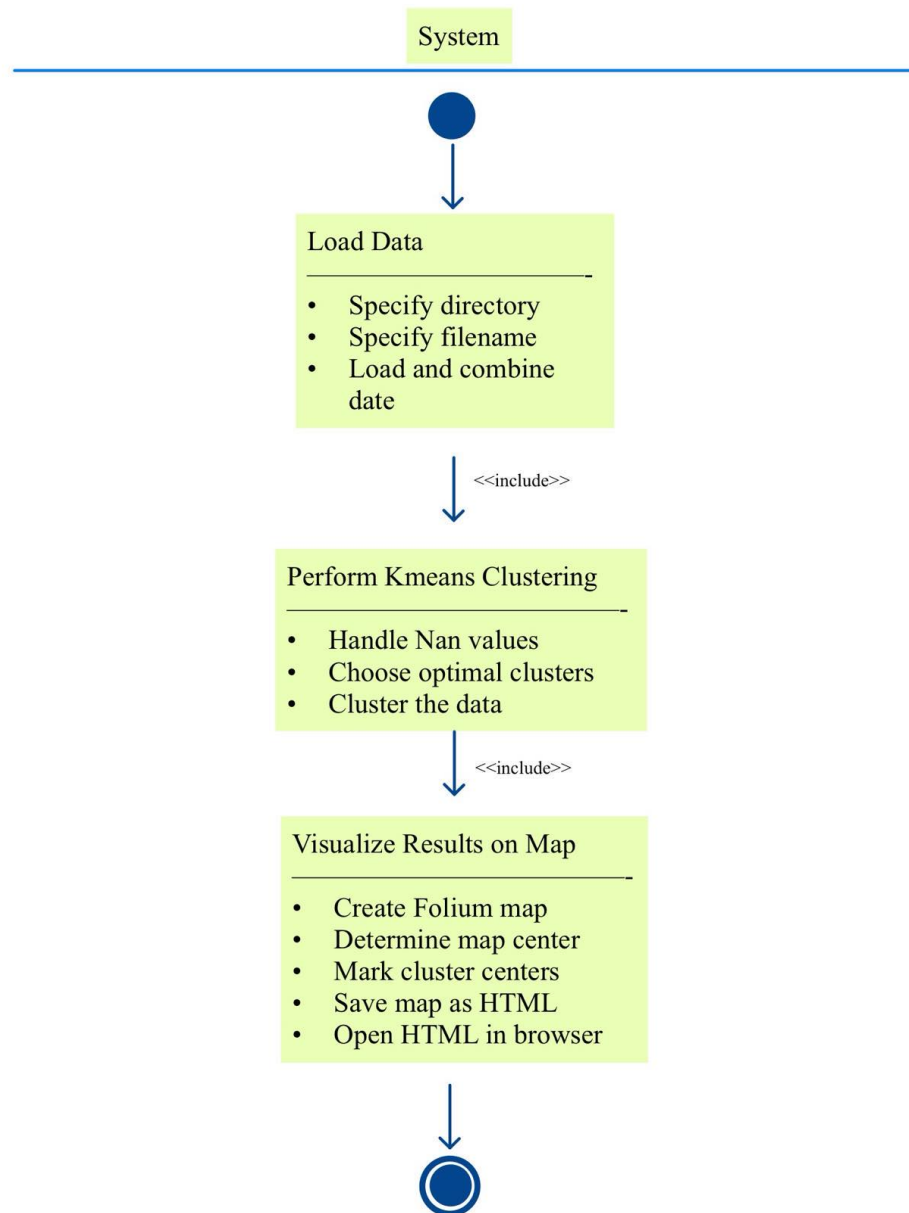


Figure 12 Use case diagram

This diagram represents the relationships between the user or external system and the three main use cases: Load Data, Perform KMeans Clustering, and Visualize Results on Map. The "<<include>>" notation signifies that the functionalities described in each subsequent use case are included within the higher-level use case.

### 3.4 Activity diagram

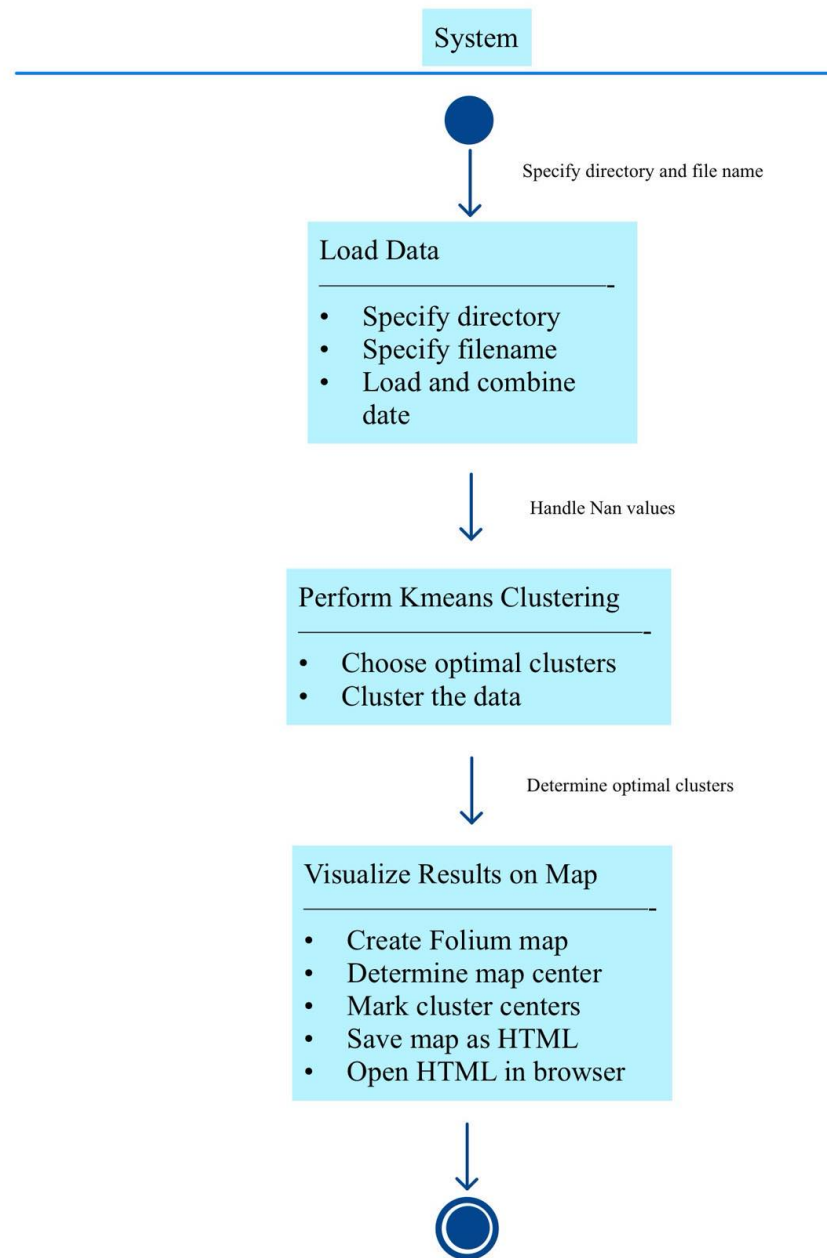


Figure 13 Activity diagram

This activity diagram representation outlines the sequence of actions within the system. The activities start from the "System" and go through the phases of "Load Data," "Perform KMeans Clustering," "Visualize Results on Map," and finally conclude with an "End" action. Each activity lists the specific tasks or operations performed as part of that phase.

## **CHAPTER 4**

### **CONCLUSION**

#### **4.1 Conclusion**

Throughout this project, extensive research was dedicated to pivotal subjects such as signal reception, methodologies for data collection, and their practical implementation. The exploration also delved into programming languages and software pertinent to this field, fostering a comprehensive grasp of their applicability.

A crucial focal point of this endeavor was the evaluation of the necessity for signal boosters for Mae Fah Luang University's wireless LAN controller network. This strategic plan was crafted with the primary objective of optimizing the existing network infrastructure.

The data collection process involved gathering diverse information dispersed by access points to an array of electronic devices, ranging from smartphones to tablets, laptops, and various other gadgets. Implemented within this framework was a database designed to automatically capture data upon device connection to a WiFi signal. This data was then transmitted to a centralized database for analysis.

#### **4.2 Discussion**

The project encountered several challenges, primarily due to its continuation from a previous senior project. One significant hurdle was understanding the code left behind by the prior team, which was essential for reconstructing the database. Additionally, aggregating user data from access points across different areas posed another obstacle. Initially, navigating various software programs posed difficulties. However, these challenges were tackled by seeking information from online sources and consulting project advisors for guidance.

Understanding the intricacies of the previous codebase and the complexities of collecting data from network broadcasters presented initial setbacks. Nonetheless, a dedicated commitment to learning, coupled with resourcefulness in seeking solutions online and leveraging the expertise of project advisors, played a vital role in overcoming these challenges. This ultimately leads to successful navigation and resolution of project complexity.



## REFERENCES

- [1] "Wireless Communications and Networking" [Online]. Available: <https://www.google.com/url?q=https://www.iqytechnicalcollege.com> [Accessed: 12.Mar.2023].
- [2] "Wireless Communication Networks and Systems". [Online]. Available: <https://books.google.co.th/books> [Accessed: 19.Mar.2023].
- [3] "Geographic Information Systems and Science". [Online]. Available: <https://books.google.co.th/books> [Accessed: 26.Mar.2023].
- [4] "Traffic Engineering Handbook". [Online]. Available: <https://books.google.co.th/books> [Accessed: 2.Apr.2023].
- [5] "Predictive Analytics for Dummies". [Online]. Available: <https://books.google.co.th/books> [Accessed: 9.Apr.2023].